

BRIDGING COURSEWORK AND PROFESSIONAL PRACTICE THROUGH A RETRIEVAL-GROUNDED, PRIVACY-PRESERVING AI LEARNING COMPANION

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Abstract – Large graduate engineering courses must balance scalability, authentic exposure to professional systems, and individualized support. While engagement with real-world professional artifacts can deepen learning, their scale and complexity often overwhelm students, and large enrollments limit instructors' ability to provide sustained guidance. Cloud-based generative AI tools further complicate this landscape by encouraging passive solution generation and raising institutional privacy concerns. To address these challenges, we present the design of a retrieval-grounded, privacy-preserving AI Learning Companion embedded within course websites to support guided exploration of instructor-curated professional codebases. The proposed system employs hybrid lexical-semantic retrieval with chunk-level indexing and a locally hosted 8B-parameter instruction-tuned language model to interpret and annotate retrieved materials rather than generate new solutions. The paper describes the system architecture, pedagogical framework, and planned evaluation methodology for future classroom deployment, arguing that locally controlled, retrieval-grounded AI systems may support scalable, authentic, and ethically aligned learning in graduate engineering education.

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1. INTRODUCTION

Teaching large, professionally oriented graduate engineering courses presents persistent challenges in balancing scalability, authenticity, and individualized support. With enrollments often exceeding one hundred students, instructors face substantial time constraints that limit opportunities for personalized guidance. Students also arrive with diverse technical backgrounds and career goals, yet scalable course structures, such as auto-graded exercises or standardized projects, leave limited room for tailored mentorship. While these approaches promote

fairness and manageable workload, they rarely reflect the open-ended, integrative, and collaborative nature of professional engineering practice.

Group projects can partially address this gap, but to remain feasible within a single academic term, they are typically tightly scoped. As a result, students rarely gain sustained exposure to how complex engineering systems evolve, integrate heterogeneous technologies, or satisfy real-world expectations for maintainability and scalability.

One promising approach to addressing this limitation is to engage students with authentic professional artifacts. In software-oriented courses, open-source projects offer access to industrial-grade, collaboratively developed codebases that embody realistic architectural decisions and documentation practices. Yet these repositories are often massive, intricate, and insufficiently documented, leaving students uncertain where to begin, what to prioritize, or how to interpret design trade-offs. Without substantial guidance, many struggle to meaningfully engage with such systems. Such engagement requires sustained guidance, but in large-enrollment classes, instructors cannot feasibly provide ongoing, individualized walkthroughs of complex systems. Without structured scaffolding, the pedagogical potential of open-source, professional systems remains underutilized in formal engineering education.

Meanwhile, the rapid rise of generative artificial intelligence (AI) has transformed how students approach technical learning. Large language models can provide immediate explanations and generate code with minimal effort, potentially lowering barriers to exploring complex systems. However, uncritical reliance on generative AI risks encouraging passive solution generation rather than deep engagement with underlying concepts. A central pedagogical dilemma therefore emerges: students who rely heavily on AI may bypass meaningful learning processes, while those who avoid AI entirely may be underprepared for contemporary engineering environments, where AI-assisted development is increasingly normative.

At the same time, integrating AI into coursework is further complicated by institutional and ethical constraints. Many commercial, cloud-based AI platforms require transmitting student queries and instructional materials to external servers. In research-intensive Canadian institutions, this raises legitimate privacy, compliance, and research ethics concerns, including adherence to Research Ethics Board (REB) guidelines and institutional data-governance policies. These constraints limit the adoption of commercial AI tools, even when pedagogically beneficial.

In response to these intertwined pedagogical and ethical challenges, this paper presents the design of a privacy-preserving AI Learning Companion embedded directly within course websites. Students interact with the Companion through a conversational interface, posing natural-language questions about system architecture, implementation strategies, or design patterns. The proposed system is designed primarily for software-oriented graduate engineering courses, where students engage with large, professionally developed codebases. Rather than generating new assignment solutions, the Companion is intended to help students navigate and interpret instructor-curated repositories containing source code, documentation, and architectural components drawn from real-world open-source projects.

Technically, the proposed Companion employs a retrieval-grounded architecture based on retrieval-augmented generation (RAG) [9] principles. Curated repositories and course materials are segmented into manageable units (e.g., functions, classes, or documentation sections) and indexed for search. When a student submits a query, the system applies a hybrid lexical-semantic retrieval strategy to identify relevant code segments and documentation, which are then interpreted and contextualized by a locally hosted 8B-parameter instruction-tuned language model constrained to retrieved materials. All indexing, retrieval, inference, and data storage are designed to occur on institutionally controlled hardware, supporting privacy, transparency, and independence from external cloud services.

This work adopts a design-oriented approach that integrates principles of experiential learning [1] and guided inquiry [14] with contemporary retrieval-augmented AI system design. Rather than reporting empirical deployment outcomes, this paper focuses on the design rationale, technical architecture, pedagogical framing, and planned evaluation methodology of the proposed system. A future pilot deployment is intended for a large graduate engineering course to examine how retrieval-grounded AI may support engagement with authentic professional artifacts in large-enrollment learning environments.

This paper makes three primary contributions:

1. **A technically grounded architecture** for locally hosted, retrieval-grounded AI assistance tailored to engineering education contexts;
2. **A pedagogical integration framework** that positions AI as a structured scaffold for engaging with authentic professional artifacts rather than as a generative shortcut;
3. **A planned evaluation methodology** for examining how privacy-preserving, retrieval-based AI systems may influence student engagement, professional reasoning, and responsible AI use in large courses.

By bridging engineering education theory with privacy-preserving AI system design, this work proposes a scalable and ethically aligned approach for integrating retrieval-grounded AI into engineering education.

2. BACKGROUND AND RELATED WORK

2.1. Authentic Professional Practice in Engineering Education

Bridging classroom instruction and professional engineering practice has long been a central objective in engineering education. Experiential learning theory emphasizes engagement with authentic tasks as critical for developing reflective and adaptive expertise [1]. Similarly, research on engineering expertise highlights the importance of open-ended problems, system-level reasoning, and integrative thinking in preparing students for professional environments [2]. Instructional frameworks in STEM education advocate incorporating real-world artifacts and professional contexts into coursework to support deeper conceptual understanding and professional identity formation [3].

In software-oriented courses, open-source software projects have been proposed as vehicles for exposing students to industrial-grade systems [4, 5]. Such repositories provide access to collaboratively developed codebases that reflect realistic architectural trade-offs and documentation practices. However, prior work also notes significant challenges, including cognitive overload, uneven documentation, and the need for substantial scaffolding to support meaningful engagement [4]. While the pedagogical value of authentic artifacts is well established, scalable mechanisms for guiding structured interaction with complex professional systems remain comparatively underdeveloped.

2.2. AI-Assisted Learning in Engineering and Computing Education

The emergence of large language models (LLMs) has accelerated research on AI-assisted learning tools in computing and engineering education. Recent work explores conversational tutors, automated feedback systems, and AI-supported programming assistance [6, 7, 12]. These tools can provide immediate explanations,

generate example solutions, and assist with debugging, potentially lowering barriers to independent exploration.

However, generative AI systems raise important pedagogical concerns. Scholars have cautioned that uncritical reliance on LLM-generated outputs may reduce opportunities for productive struggle and deep conceptual engagement [6, 8]. Generative models can also produce hallucinated or inconsistent responses, complicating their use in formal instructional contexts [8]. Additionally, many AI-assisted educational systems rely on commercial cloud infrastructures, limiting instructor control over data handling and model behavior.

Retrieval-augmented approaches, which ground AI responses in curated materials rather than relying solely on generative outputs, have emerged as a strategy for improving factual reliability and contextual relevance [9]. While retrieval-augmented generation has gained attention in broader AI research, its application in engineering education remains relatively limited [16]. Few documented systems explicitly integrate retrieval-grounded AI with structured engagement with authentic professional artifacts.

2.3. Responsible AI and Data Governance in Educational Contexts

Alongside pedagogical debates, ethical considerations surrounding AI deployment in education have received increasing attention [13, 15]. Holmes et al. emphasize the need for transparency, accountability, and human oversight in educational AI systems [10]. Broader analyses of generative AI in higher education similarly underscore risks related to bias, misinformation, and reduced learner autonomy [8].

In research-intensive institutional contexts, data governance and privacy frameworks impose additional constraints. The Tri-Council Policy Statement (TCPS 2) in Canada establishes guidelines for protecting participant data and minimizing privacy risks in research and educational settings [11]. Commercial AI platforms typically require transmitting user queries and instructional materials to external servers, where data storage, retention, and secondary use practices may not be fully transparent. These constraints complicate the integration of cloud-based AI systems into coursework and classroom-based research.

Despite widespread discussion of responsible AI principles, comparatively limited literature addresses the design of locally hosted, privacy-preserving AI infrastructures tailored specifically for engineering education contexts.

2.4. Research Gap

Existing scholarship establishes three converging strands: (1) the pedagogical value of engaging students with authentic professional artifacts, (2) the rapid

adoption of generative AI tools in computing and engineering education, and (3) the ethical importance of responsible, privacy-aware AI deployment. However, these strands have rarely been integrated.

Current AI-assisted systems in engineering education have largely emphasized generative support tasks such as solution production, while comparatively less attention has been devoted to helping students systematically interpret and navigate curated professional systems. Conversely, educational efforts that incorporate authentic professional artifacts into coursework often rely on substantial instructor guidance and therefore remain difficult to scale. At the same time, privacy-preserving and institutionally controlled AI architectures continue to be underexplored in educational practice, despite growing concerns regarding data governance, intellectual property, and responsible AI use.

This paper addresses this gap by presenting the design of a locally hosted, retrieval-grounded AI Learning Companion intended to support guided inquiry into instructor-curated professional repositories. By integrating hybrid lexical-semantic retrieval with a pedagogically grounded framework for engaging authentic systems, this work proposes a scalable and ethically aligned approach to AI-assisted learning in graduate engineering education.

3. SYSTEM DESIGN

3.1. System Overview

The AI Learning Companion is designed as a locally hosted, retrieval-grounded conversational system embedded directly within the course website. Students interact with the Companion through a dedicated “Ask AI Companion” interface, submitting natural-language questions about system architecture, implementation strategies, or software design patterns. The interface is intentionally lightweight and integrated into the existing course platform to minimize friction and normalize responsible AI use as part of routine learning activity.

Unlike generative AI tools that produce new assignment solutions, the Companion is designed to guide students through instructor-curated professional repositories. These repositories consist of structured collections of source code, documentation, configuration files, and architectural components drawn from real-world open-source projects aligned with course objectives.

The system adopts a retrieval-augmented architecture. When a student submits a query, the Companion retrieves relevant segments from indexed repositories and course materials, then generates a contextualized explanation grounded explicitly in those retrieved sources. This design constrains responses to instructor-curated materials and reduces the likelihood of hallucinated or unverified content. Rather than simply returning isolated documents or code segments through keyword matching alone, the Companion synthesizes contextual explanations across

multiple retrieved artifacts. This allows the system to explain architectural relationships, implementation trade-offs, and conceptual connections in addition to locating relevant implementation details.

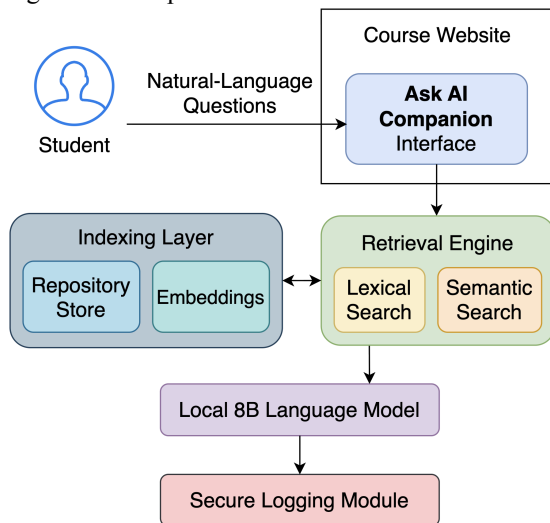


Figure 1: System architecture.

As shown in Figure 1, the overall system architecture consists of five primary components:

1. A web-based front-end interface embedded within the course website;
2. A back-end retrieval engine implementing hybrid lexical-semantic search;
3. An indexing layer storing segmented repository content, embeddings, and metadata;
4. A locally hosted instruction-tuned language model for explanatory generation;
5. A secure logging module for anonymized interaction analytics.

At runtime, a student query flows through these components sequentially: the retrieval engine queries the indexing layer to identify relevant segments, constructs a structured context window, and passes this context to the local language model for constrained explanatory generation. Interaction metadata are stored in anonymized form for aggregate analysis and system refinement.

All components are deployed on institutionally controlled hardware to ensure privacy and governance compliance.

3.2. Retrieval and Indexing Architecture

Curated Repository Selection and Content Scope

The Companion operates exclusively on instructor-curated materials. In software-oriented courses, curated repositories are selected based on pedagogical relevance and quality criteria, including alignment with course learning objectives, architectural clarity, documentation completeness, community adoption indicators (e.g., active maintenance and substantial GitHub engagement), and

representativeness of professional design practices. Rather than indexing arbitrary public repositories, the system relies on a deliberately bounded corpus chosen to support specific instructional goals.

In addition to open-source codebases, the indexed corpus may include course materials such as lecture slides, assignment handouts, architectural diagrams, and instructor-written examples. Integrating both professional repositories and course-specific materials allows the Companion to connect real-world implementations with formal instructional content.

Chunk-Level Indexing

Curated repositories are segmented into manageable units, including:

- Individual functions and methods
- Class definitions
- Documentation sections
- Configuration files
- Lecture slides or annotated instructional notes

Each segment is indexed independently to enable fine-grained retrieval. Metadata fields include file path, repository origin, module category, content type (e.g., code vs. documentation vs. instructional material), and topical tags aligned with course concepts. This structured segmentation supports precise retrieval while preserving contextual traceability to the original source.

Hybrid Retrieval Strategy

When a student submits a natural-language query, the system applies a hybrid retrieval strategy that combines two complementary search mechanisms.

1. **Lexical retrieval (BM25-based ranking)** identifies repository segments containing exact or closely matching keywords from the query. This method is particularly effective for locating specific API names, function calls, configuration terms, or domain-specific terminology.
2. **Semantic retrieval (embedding similarity search)** represents both the query and repository segments as high-dimensional vectors capturing their contextual meaning. This allows the system to identify conceptually related content even when exact keywords do not match (e.g., linking “user authentication” with code implementing token validation).

The scores from lexical and semantic retrieval are combined using weighted aggregation to produce a ranked list of candidate segments. This hybrid approach improves robustness: lexical search ensures precision for technical terms, while semantic search supports conceptual interpretation of broader questions. Together, they reduce the likelihood of missing relevant content or retrieving superficially similar but conceptually unrelated segments.

Context Construction

Top-ranked segments are assembled into a structured context window that is passed to the language model along with system-level instructions constraining its behavior. The model is prompted to generate explanatory responses explicitly grounded in the retrieved materials. Specifically, it is instructed to:

- Identify and describe the purpose of relevant functions, classes, or modules within the broader system architecture.
- Explain how specific implementation choices (e.g., authentication mechanisms, concurrency handling) relate to established design patterns.
- Highlight trade-offs reflected in the code (e.g., modularity vs. performance, abstraction vs. simplicity).
- Link retrieved implementation examples to course material.

For example, if a student asks how authentication is handled in a distributed system, the model retrieves token-validation functions and associated middleware components, then explains how these components implement session management patterns covered in class.

Importantly, the model is explicitly instructed not to generate assignment solutions, rewrite student code, or produce new implementations. Its role is limited to interpreting and contextualizing instructor-curated examples.

3.3. Model Selection and Deployment

The proposed Companion architecture employs an 8B-parameter open-source instruction-tuned language model. Instruction tuning refers to a post-pretraining process in which models are fine-tuned to follow task-level prompts and generate structured, context-aware responses. This characteristic is essential for the Companion, which is designed to produce explanatory summaries grounded in retrieved materials while adhering to explicit constraints (e.g., avoiding generation of new assignment solutions).

The 8B-parameter model was selected to balance explanatory quality, inference latency, and local hardware constraints. While larger models may offer improvements in generative fluency, the Companion's retrieval-grounded design reduces the need for expansive parametric knowledge. Because responses are constrained to instructor-curated materials, the model primarily performs structured interpretation and contextualization rather than open-ended synthesis. In preliminary prototyping, larger models were observed to increase inference latency without clear proportional gains in explanatory usefulness for the intended educational setting.

The proposed deployment environment consists of a Mac mini configured with 32GB of memory and a dedicated 4TB solid-state storage device (SSD). The SSD is intended to store:

- **Local copies of language models**, supporting offline inference and eliminating dependence on external API services.
- **Precomputed embedding vectors** for indexed content, enabling efficient semantic similarity search without recomputing embeddings at query time.
- **Indexed repository content and metadata**, allowing low-latency lexical and hybrid retrieval across curated materials.
- **Anonymized interaction logs** intended for future aggregate analytics while preserving student privacy.

Separating storage-intensive components (models, embeddings, indexed corpora) onto a dedicated SSD is intended to improve input/output performance and reduce memory pressure during concurrent usage. Similarly, precomputing embeddings and maintaining a persistent search index are expected to reduce response latency and support classroom-scale deployment without reliance on external cloud infrastructure.

By restricting the model's role to retrieval-grounded summarization rather than open-ended generation, computational load may also be reduced. This architectural constraint is intended not only to improve response stability and inference efficiency but also to support pedagogical reliability by limiting outputs to instructor-curated materials.

3.4. Privacy and Data Governance Design

A central design principle of the Companion is privacy preservation through institutional control. All indexing, retrieval, inference, and storage occur on departmental hardware within the university network. No student queries or course materials are transmitted to external servers.

Interaction logs are anonymized at the point of storage and used exclusively for aggregate analytics, including:

- Identifying frequently queried architectural topics
- Detecting areas of conceptual confusion
- Refining repository curation

This local deployment architecture aligns with institutional privacy standards and Canadian research ethics guidelines. By eliminating dependence on commercial API services, the system provides instructors with full transparency over data handling and retrieval sources, supporting responsible AI integration in educational contexts.

4. PEDAGOGICAL FRAMEWORK

While Section 3 described the technical architecture of the proposed AI Learning Companion, the pedagogical value of the system depends fundamentally on its educational grounding. This section articulates the educational principles guiding the system's design and clarifies how its retrieval-grounded architecture is intended to support inquiry-driven engagement with

authentic professional artifacts. Drawing on experiential learning, guided inquiry, and responsible AI integration, the Companion is positioned not as an automated solution generator but as a structured interpretive scaffold within large-scale graduate engineering instruction.

4.1. Guided Inquiry and Experiential Learning

The AI Learning Companion is grounded in principles of guided inquiry and experiential learning. Experiential learning theory emphasizes iterative engagement with authentic tasks, reflection, and conceptual integration [1]. Rather than presenting simplified or decontextualized examples, the Companion directs students to interact with professionally developed systems that embody real-world complexity. However, experiential exposure alone is insufficient; without scaffolding, large and unfamiliar systems can produce cognitive overload rather than productive learning.

Guided inquiry frameworks highlight the importance of structured exploration in which learners are supported in asking questions, identifying relevant information, and constructing meaning [14]. The proposed Companion operationalizes guided inquiry by enabling students to pose natural-language questions about architectural components, implementation strategies, and design trade-offs. Instead of providing final answers detached from context, the system retrieves curated artifacts and generates explanatory commentary that makes expert reasoning visible. In this way, AI serves as a scaffold for inquiry rather than a substitute for reasoning.

4.2. Scaffolding Engagement with Authentic Systems

Authentic professional artifacts, such as large open-source repositories, expose students to realistic engineering constraints, including modularity decisions, documentation practices, and system evolution. However, the scale and heterogeneity of such artifacts often exceed what can be productively explored within a single academic term.

The proposed Companion is intended to address this challenge by combining instructor curation with fine-grained retrieval. Instructor selection narrows the corpus to pedagogically aligned repositories, while chunk-level indexing allows students to examine specific components without losing traceability to the broader system. This structure supports a form of “bounded authenticity,” where students engage with real systems within a guided and interpretable scope.

By grounding explanations in retrieved segments, the system also models how professional engineers navigate large codebases — not by memorizing entire systems, but by searching, interpreting, and reasoning across components.

4.3. Positioning AI as Interpretive Support Rather Than Generative Shortcut

A central pedagogical concern surrounding generative AI is the risk of dependency and passive solution consumption. When AI tools generate complete answers or code solutions, they may reduce opportunities for productive struggle and conceptual consolidation.

The proposed Companion is designed to mitigate this risk via architectural constraints. Responses are restricted to retrieved, instructor-curated materials, preventing the system from generating assignment solutions or new implementations. Instead, it performs interpretive synthesis by explaining, contextualizing, and connecting existing examples to course concepts. This shifts AI’s role from solution generator to explanatory mediator.

By limiting generation to retrieval-grounded contexts, the system aligns AI assistance with instructional goals. Students remain responsible for applying concepts, designing solutions, and completing assignments, while the Companion supports interpretation of professional artifacts and clarification of architectural reasoning.

4.4. Instructor-Guided Knowledge Boundaries

Unlike commercial AI platforms trained on broad and opaque datasets, the Companion is designed to operate within clearly defined knowledge boundaries established by the instructor. Repository selection, content segmentation, and tagging decisions are treated as pedagogical design choices rather than purely technical processes. This curation process helps ensure that AI-generated explanations remain grounded in course-relevant materials and professional practices aligned with instructional objectives.

Because the system relies on instructor-curated repositories, instructors retain visibility into the materials used to generate responses. This supports greater transparency, pedagogical alignment, and instructional control than commercial generative AI systems, where training data and retrieval sources are often inaccessible or poorly defined.

Future anonymized interaction analytics may also support iterative course refinement. Aggregate query patterns could help identify conceptual bottlenecks, frequently misunderstood architectural elements, or topics requiring additional instructional emphasis. In this way, the proposed system can support not only student inquiry but also ongoing refinement of instructional materials and repository curation.

5. PLANNED EVALUATION METHODOLOGY

The proposed AI Learning Companion is intended for future deployment in a professionally oriented graduate-level engineering course. As a design-based, practice-oriented intervention, the planned evaluation

seeks to examine how retrieval-grounded, locally hosted AI support influences students' engagement with authentic professional systems, patterns of inquiry, and perceptions of responsible AI use. More broadly, the evaluation aims to understand how retrieval-grounded AI can function as a pedagogical scaffold within an authentic instructional context.

5.1. Planned Deployment Context

The proposed pilot deployment is intended for a graduate engineering course focusing on modern software systems and cloud-based architectures. Students in the course engage in assignments and group projects requiring architectural reasoning, system integration, and critical analysis of real-world design patterns.

Following implementation, the Companion is designed to be integrated into the course website and introduced during a scheduled lecture session. Students would be informed of the system's intended pedagogical role through official course communication. Use of the system would remain entirely voluntary.

Future evaluation activities, including surveys, interaction analytics, and voluntary interviews, would be conducted following institutional research ethics approval.

5.2. Research Questions

The evaluation is structured around three primary research questions intended to guide future evaluation activities:

RQ1. To what extent does access to the Companion support students' ability to navigate and interpret large professional codebases?

This question will be examined through indicators such as:

- Self-reported confidence in locating and understanding relevant architectural components within large repositories.
- Students' descriptions of how they trace relationships between modules, APIs, or configuration layers.
- Reported changes in strategies for exploring unfamiliar codebases.

RQ2. How do students perceive retrieval-grounded AI compared to general-purpose generative AI tools in supporting responsible and meaningful learning?

This question will be explored through:

- Survey items contrasting explanatory clarity versus solution generation.
- Student reflections on whether the Companion supports understanding versus task completion.
- Perceived differences in trust, transparency, and relevance between retrieval-grounded responses and open-ended generative outputs.

RQ3. How is the Companion used in practice within a large engineering course?

This question will draw on:

- Usage frequency and temporal patterns (e.g., steady exploration versus deadline-driven spikes).
- Categorization of query types (e.g., architectural clarification, design pattern explanation, debugging interpretation).
- Qualitative accounts of how students integrate Companion responses into project work.

These indicators aim to capture not only perceived confidence but also patterns of inquiry and engagement that reflect structured interaction with authentic professional systems.

5.3. Data Sources and Instruments

The evaluation will draw on three complementary data sources:

1. Pre- and Post-Deployment Surveys

Students will be invited to complete voluntary, anonymized surveys at the beginning and end of the term. Survey instruments will measure:

- Confidence in navigating large codebases
- Perceived ability to analyze architectural decisions
- Attitudes toward AI-assisted learning
- Perceived differences between generative and retrieval-grounded AI tools

Likert-scale items and short open-ended responses will be included to capture both quantitative shifts and qualitative reflections.

2. Interaction Analytics

Anonymized interaction logs will be analyzed to examine:

- Frequency of usage over time
- Distribution of query types (e.g., architectural clarification, design pattern explanation, implementation-level inquiry)
- Temporal patterns relative to assignment deadlines

These analytics will provide behavioral evidence of how students use the Companion in authentic contexts.

3. Voluntary Focus Group Interviews

A small number of volunteer students may be invited to participate in semi-structured interviews to explore:

- How they integrate Companion responses into their workflow
- Whether the system supports deeper reasoning or task acceleration
- Perceived limitations of retrieval-grounded AI

Interview data will support interpretive analysis of student experience beyond survey metrics.

5.4. Analysis Plan

Quantitative survey data will be analyzed using descriptive statistics and paired comparisons where appropriate to identify shifts in self-reported confidence and AI perceptions over the term.

Interaction analytics will be examined to identify recurring inquiry themes and to detect potential over-reliance patterns (e.g., repeated near-deadline spikes in usage).

Qualitative survey responses and interview transcripts will be coded using thematic analysis to identify recurring perceptions of scaffolding, professional insight, or dependency. Together, these analyses aim to triangulate perceived learning impact, behavioral engagement patterns, and qualitative experiences.

5.5. Ethical Considerations

Any future evaluation involving student participants will be conducted in accordance with institutional research ethics policies. Research ethics approval for the planned study has been obtained from the University of Toronto Research Ethics Board.

Use of the Companion and participation in any associated surveys or interviews will remain entirely voluntary and will not affect students' course standing. Any interaction logs used for analysis will be anonymized at the point of storage, and no personally identifiable information will be included in the analysis.

Because the Companion is designed to operate entirely on institutionally controlled hardware and does not transmit student queries or course materials to external servers, the proposed architecture is intended to reduce privacy and data-governance risks relative to commercial cloud-based AI platforms.

6. DISCUSSION, LIMITATIONS, AND FUTURE WORK

The AI Learning Companion represents a design-based effort to integrate retrieval-grounded AI with guided inquiry in graduate engineering education. While the system architecture, pedagogical framework, and evaluation methodology have been designed, classroom deployment and formal evaluation remain ongoing. Accordingly, this paper focuses on design rationale, system architecture, pedagogical framing, and planned evaluation rather than empirical classroom findings.

Several limitations warrant consideration. First, the effectiveness of the Companion may depend heavily on instructor curation. The quality, representativeness, and documentation of selected repositories influence retrieval accuracy and explanatory clarity, while maintaining curated corpora requires ongoing instructional effort. Second, although hybrid retrieval improves robustness, embedding-based similarity search may still retrieve contextually related but pedagogically suboptimal

segments, particularly in heterogeneous codebases. Third, while retrieval-grounded constraints reduce hallucination and solution generation, they do not eliminate the risk of student over-reliance. Hence, clear instructional framing remains necessary to position the Companion as interpretive support rather than task automation.

Future work includes pilot deployment, collection of survey and interaction data, and qualitative investigation of how students engage with retrieval-grounded AI support in practice. Additional directions include studying longitudinal effects, comparing the system with cloud-based generative AI tools, and adapting the framework to non-software engineering domains.

Despite these limitations, the Companion contributes a replicable model for locally controlled, privacy-preserving AI integration in engineering education. By combining hybrid retrieval with instructor-curated knowledge boundaries, the proposed system offers a scalable and ethically grounded approach to supporting student engagement with authentic professional artifacts.

7. CONCLUSION

This paper presented the design and pedagogical grounding of a privacy-preserving AI Learning Companion for large graduate engineering courses. The proposed system combines hybrid lexical-semantic retrieval with instructor-curated professional repositories to support guided exploration of real-world software systems. Rather than generating assignment solutions, the Companion is designed to interpret and contextualize retrieved materials, positioning AI as a scaffold for inquiry rather than a substitute for student reasoning.

By integrating retrieval-grounded AI, instructor-curated knowledge boundaries, and locally controlled infrastructure, this work proposes a scalable and ethically aligned approach to supporting authentic engagement with professional engineering artifacts in graduate engineering education.

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